

What One Aggregate Equation Can Identify About Credit-Conditioned Consumption (Online Appendix)

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1 Online Appendix to [manuscript title TBD]

Online appendix — not for print.

This appendix supports the main text with (A) full data-construction detail, (B) the complete twelve-specification battery, (C) the robustness battery, (D) the credit-conditions-index (CCI) construction and placebo battery, (E) the permanent-income construction, and (F) the structural-parameter inference battery, including a bounds/stationarity test and a nested bootstrap. Tables are numbered within each section (Table A1, A2, ... ; Table B1, B2, ... ; and so on) and every table records its source file(s) in the note beneath it.

Contents

- A. Data construction detail
- B. The twelve-specification battery
- C. Robustness battery
- D. Credit-conditions-index construction and placebo battery
- E. Permanent-income construction
- F. Structural-parameter inference: bounds test, ablation, Spec-11 placebo, and the nested bootstrap

1.1 A. Data construction detail

1.1.1 A.1 Splice methodology — growth-rate chain-linking

For each adjacent pair of source layers, the splice anchors the level at the first quarter where both series are non-missing, then back-casts using the base series' own quarter-on-quarter growth rates:

```
chained[t] = overlay[t_anchor] × (base[t] / base[t_anchor])   for t < t_anchor
chained[t] = overlay[t]                                         for t ≥=
t_anchor
```

This is the standard ABS chain-linking convention, used throughout in preference to a mean-ratio splicing convention: it preserves the base series' growth rates exactly and pins the level to the overlay series at the join, so there is no level discontinuity at any join quarter — including at the long TRYM/legacy house-price overlap and the bridge/current house-price overlap, where a mean-ratio convention would produce step jumps.

The relative house-price-to-income ratio used in estimation is

$$\ln_hp_over_y = \log(hpi \times pop_millions / ydi_ann_nom)$$

i.e. the log of the nominal house-price index divided by nominal annualised disposable income per capita. Because numerator and denominator are both nominal, the consumption deflator cancels exactly and the ratio is identical to the real house price over real income per capita:

$$(hpi/defl) / (ydi_ann_nom/pop/defl) = hpi \times pop_millions / ydi_ann_nom$$

This nominal/nominal (equivalently real/real) construction is the one used throughout, in preference to a nominal/real construction: mixing a nominal numerator with a real denominator (or vice versa) would leave the economy-wide price level inside the ratio (correlation 0.98 with the consumption deflator), so the term would measure aggregate inflation rather than relative house-price pressure. The nominal/ nominal construction avoids this by construction, since the deflator cancels exactly.

1.1.2 A.2 Credit conditions index — data sources

See §D for the full CCI construction and placebo battery. The two observable/constructed proxies are: `cci_ratio` (log of housing-credit flow, ABS Cat 5601.0, divided by an eight-quarter moving average of nominal disposable income, available from 2002Q3) and `cci_williams` (the smoothed-step spline described in §D). The first-home-buyer share `fhb_share` = `fhb_loans` / `total_new_loans` is also constructed from ABS Cat 5601.0 from 2002Q3.

1.1.3 A.3 RBA D-tables (pre-1988 back-extension)

Three RBA historical statistical tables support the pre-1988 portion of the back-extended sample.

D03 — M3 (monetary aggregate). `m3_aggregate` is loaded from `d03hist.xlsx`, series `DMAM3N` (M3, original/not-seasonally-adjusted, \$ billion): total economy-wide currency plus transaction deposits plus all other deposits at ADIs plus ADI-issued certificates of deposit. Coverage: monthly, 1959Q3–2026Q1, aggregated to quarterly by the mean of the three monthly observations. `DMAM3N` carries a definitional series break at August 1976, where the level jumps by +14.25 log per cent month-on-month against a mean monthly log change of 0.8 log per cent over the full history. The break falls inside the opening quarters of the

1976Q3+ back-extended spine: it makes the 1976Q4 quarterly log-difference of M3 (+9.5 log per cent) an outlier of roughly 4.8 standard deviations of the series' quarterly-growth distribution over the spine window, and the outlier propagates into every M3-based back-extension proxy (`m3_household_proxy`, `fin_deposits_proxy`, `nla_y_proxy`, `networth_y_proxy`) at the deepest end of the back-extended sample. Back-extension results that lean on the first spine quarters should be read with this break in mind. M3 is the headline liquid-asset proxy for the pre-1988 portion of the sample, where ABS sectoral household-balance-sheet deposits are not available.

D02 — total credit. `credit_total_d02` is constructed by growth-rate-splicing two RBA D02 series across the July 2019 RBA conceptual reform: `DLCACN` (total credit, original, \$ billion) for 1976Q3–2019Q2, and `DLCACSFN` (total credit including select financial businesses, the post-2019 successor) for 2019Q3 onward. The two series have no quarterly overlap, so the splice anchors levels at the boundary (`first_post × pre[t] / last_pre`); the join is continuous in level but the implicit growth rate at the boundary is exactly zero (there is no overlap to estimate it from). This is total credit, not housing-specific; the housing-specific D02 series (`DLCAC0HN`, `DLCACIHN`) only extends back to 1990Q1 in the current vintage.

D01 — growth in selected financial aggregates. Downloaded as `d01hist.xlsx` (monthly growth rates of the aggregates whose levels are in D02). Not needed for the current back-extension, since D02 already extends to 1976Q3, and is retained for future use.

1.1.4 A.4 Pre-1978 labour force

A compiled historical labour-force dataset (188 quarterly rows, 1964Q3–2011Q2; sources: ABS Cat 6204.0 historical labour force (archived), the ABS Year Book Australia, the Foster (1996) *Australian Economic Statistics 1949–50 to 1996–97* compilation, and RBA Occasional Paper No 8) provides total resident population, working-age population (15–64), civilian labour force, and unemployed persons, all in thousands; conceptually consistent with the current ABS Cat 6202.0 series (which begins February 1978) at the join.

The historic series are growth-rate-spliced onto the modern (1978+) series at 1978Q1: `pop_15_64` → `pop_millions` (the working-age population denominator used in per-capita normalisation), `labour_force` to the master `labour_force`, and `unemployed/labour_force × 100` to the master `unemp_rate` (level replacement before 1978Q1). This splice gives six quarters, 1976Q3–1977Q4, non-missing values for all per-capita and labour-force-derived variables — the binding remaining constraint for the 1976Q3 back-extension, since consumption, income, mortgage rate, house prices, M3, total credit and the prime-age share already extend to 1976Q3 on the raw download.

1.1.5 A.5 Household-allocated M3 (`m3_household_proxy`)

The pre-1988 liquid-asset proxy allocates M3 to the household sector via the wage share of GDP:

$$\text{m3_household_proxy} = \text{m3_aggregate} \times \text{wage_share} / 100$$

The wage share (compensation of employees / GDP) is a documented simplification for the back-extension: it captures most of household factor income but omits mixed income and property income receivable (together roughly ten percentage points, tracking wage share over time). `wage_share` is taken from ABS Cat 5206.0 Table 24 (analytical series A2302604K, 1959Q3–2024Q4; range 49–62 per cent over the sample, from ~60 per cent in the 1970s to

~50 per cent today). Coverage of `m3_household_proxy`: 1976Q3+, values \$22bn (1976Q3) to \$1,673bn (2024Q4).

1.1.6 A.6 Aggregate net-worth proxy (`networth_y_proxy`)

To enable the aggregate-net-worth specifications to fit on the back-extended sample, an aggregate net-worth proxy combines M3 allocated to households with an `hpi × pop_millions` back-cast of housing wealth, then growth-rate-splices that raw aggregate ratio onto the official `networth_y` at 1988Q3 so that the proxy equals the official series from 1988Q3 onward and back-casts smoothly through 1976Q3:

```
housing_wealth_proxy[t] = housing_wealth[1988Q3] × (hpi[t]/hpi[1988Q3])
                        × (pop_millions[t]/pop_millions[1988Q3])    for t <
1988Q3
                        = housing_wealth[t]                          for t >=
1988Q3

raw_proxy[t] = (m3_household_proxy[t] × 1000 + housing_wealth_proxy[t]) /
ydi_ann_nom[t]
scale        = networth_y[1988Q3] / raw_proxy[1988Q3]
networth_y_proxy[t] = networth_y[t]          for t >= 1988Q3
                  = raw_proxy[t] × scale      for t < 1988Q3
```

The $\times 1000$ term converts `m3_household_proxy` (\$ billion, RBA D03 units) onto the \$ million scale of `housing_wealth_proxy` and `ydi_ann_nom`; without it the M3 term contributes roughly 0.01 per cent of the numerator, rather than its intended ~12 per cent, and the aggregate proxy collapses to a housing-only back-cast.

Caveats: the back-cast omits equities and superannuation (a small household asset class pre-1992) and omits debt netting (mortgage debt was a much smaller balance-sheet share in the 1970s than today); it is used only for back-extension exercises, never as a substitute for the official `networth_y` on the modern sample. Proxy values at key dates: 5.05× annual income (1976Q3), 4.71 (1980Q1), 4.78 (1985Q1), 5.37 (1988Q3, anchored to the official series), 10.19 (2024Q4) – consistent with the historical Australian wealth-to-income path (flat through the 1970s/early 1980s, sharply rising post-1985 financial deregulation).

1.1.7 A.7 Disaggregated wealth proxies

Four further proxies extend the disaggregated wealth components to the back-extended 1976Q3+ sample. Each equals the official series for $t \geq 1988Q3$; for $t < 1988Q3$ it back-casts via the most relevant available aggregate.

- **ha_y_proxy** = `housing_wealth_proxy` / `ydi_ann_nom` (§A.6 back-cast). Values: 2.68 (1976Q3) → 2.83 (1988Q3) → 6.41 (2024Q4).
- **fin_deposits_proxy** anchors `fin_deposits[1988Q3]` and grows by `m3_household_proxy`.
- **fin_loans_proxy** anchors `fin_loans[1988Q3]` and grows by `credit_total_d02` (§A.3).
- **nla_y_proxy** = (`fin_deposits_proxy` – `fin_loans_proxy`) / `ydi_ann_nom`. Values: +0.20 (1976Q3, households net liquid creditors) → -0.05 (1988Q3) → -0.72 (2024Q4, modern net-debtor position). The sign flip around 1988 captures the post-deregulation debt build-up – precisely the variation the credit-conditions channel is meant to explain but which the modern (1988Q3+) data window largely excludes.

- **eq_y_proxy** is held constant at its 1988Q3 value pre-1988 (Australian household equity holdings were a small wealth share in the late 1970s/early 1980s). Value: 0.60 (constant) → 0.92 (2024Q4).
- **super_y_proxy** is a linear ramp from 0.1× the 1988Q3 value at 1976Q3 to the 1988Q3 value, then official thereafter (the 0.1 anchor matches Williams' (2010) Table A.1 ballpark for the pre-Superannuation-Guarantee era). Values: 0.07 (1976Q3) → 0.66 (1988Q3) → 2.44 (2024Q4).

These proxies are adequate for testing whether sample length is the binding constraint on identifying the LIVES credit channels (§C.14–C.16 below show it is not) but should not be over-interpreted as reconstructions of the 1970s household balance sheet: the equities proxy is a constant-share assumption, the superannuation proxy a linear ramp, and the deposits/debt proxies are aggregate-growth back-casts rather than household-sector measurements. All headline results use the **official** 1988Q3+ disaggregated series ($n = 146$ full, $n = 126$ pre-COVID); the proxies are confined to back-extension robustness exercises.

Coherence check. At the 1988Q3 boundary, official `networth_y` (broad, ABS closing net worth) = 5.37; the aggregate `networth_y_proxy` = 5.37 by construction; the sum-of-disaggregated `networth_y_disagg_proxy` (`ha_y_proxy` + `nla_y_proxy` + `eq_y_proxy` + `super_y_proxy`) = 4.04. The roughly 25 per cent gap is the "other wealth" component of ABS closing net worth (life-office reserves, unincorporated business equity, etc.) absent from the narrow definition.

1.1.8 A.8 Master variable coverage tiers

Table A1 — Master variable coverage under the 1976Q3+ spine. Source: model dataset build log.

First non-NA	n	Variables (selected)
1976Q3	~56	<code>cons</code> , <code>ydi</code> , <code>hpi</code> , <code>mortgage_rate</code> , <code>M3</code> , <code>total credit</code> , <code>prime_age_share</code> , <code>all dummies</code> , <code>m3_household_proxy</code> , <code>ha_y_proxy</code> , <code>nla_y_proxy</code> , <code>eq_y_proxy</code> , <code>super_y_proxy</code> , <code>networth_y_proxy</code> , <code>ln_networth_y_proxy</code> , <code>cci_kalman</code> , <code>cons_real_pc</code> , <code>ydi_real_pc</code> , <code>npv_real_pc</code> , <code>labour_force</code> , <code>unemp_rate</code> , <code>lf_share</code> , <code>pop_millions</code> , <code>ln_cons_real_pc</code> , <code>ln_ydi_real_pc</code> , <code>ln_hp_over_y</code> , <code>ecm_lag</code> (1976Q4)
1977Q3	2	<code>real_rate</code> , <code>hicp_4q_ann</code> (4-quarter CPI lag)
1978Q2	1	<code>ydi_ann_8qma</code> (8-quarter MA)
1988Q3	~21	official disaggregated wealth (<code>ha_y</code> , <code>eq_y</code> , <code>super_y</code> , <code>ilfa_y</code> = <code>eq_y</code> + <code>super_y</code> , <code>nla_y</code> , <code>networth_y</code> , <code>debt_y</code>), <code>housing_wealth</code> , <code>fin_deposits/equities/super/loans</code> (ABS 5232) — $n = 146$
2002Q3	5	<code>cci_ratio</code> , <code>fhb_share</code> , <code>housing_loan_flow</code> , <code>fhb_loans</code> , <code>non_fhb_loans</code> — $n = 90$
2009Q1	2	<code>mortgage_interest_burden_rba</code> , <code>mortgage_payment_burden_rba</code> (RBA E13) — $n = 64$

The 1988Q3 tier binds the faithful LIVES specification ($n = 146$ full, $n = 126$ pre-COVID); the 2002Q3 tier binds the conventional baseline ($n = 86$ full, $n = 66$ pre-COVID). The combined

illiquid financial ratio `ilfa_y` first becomes available at 1988Q3, alongside its `eq_y` and `super_y` constituents.

1.2 B. The twelve-specification battery

Table B1 — Specification number → manuscript name (the three named specifications).

Spec	Manuscript name	Role
6	Conventional constant-MPC disaggregated ECM	Baseline (retained for comparability with prior literature)
11	Faithful LIVES specification	Headline (narrative lead)
12	Williams-calibration-imposed LIVES	Negative control (shows what fails to transfer)

The ladder also includes a long-history variant (Spec 6b) and a measured-burden variant (Spec 7b), giving fourteen fitted forms in total across the two named "6/6b" and "7/7b" pairs plus Specs 1–5 and 8–12.

1.2.1 B.1 The fourteen-specification ladder

Table B2 — The specification ladder. Each step adds one structural element: Specs 1–3 are aggregate-net-worth error-correction models; Specs 4–7b disaggregate wealth and add cohort/burden terms; Specs 8–9 introduce the CCI interactions (spline and Kalman extractions); Specs 10–12 are the Williams-aligned forms.

Spec	Description	Long-run regressors / notes
1	Aggregate net worth	ln_networth_y, ln_hp_over_y, real_rate, ln_yp_over_y, ecm_lag
2	Spec 1 + short-run CCI	adds $\Delta^2 \log$ CCI lag 2 to short-run set
3	Net worth in levels	replaces ln_networth_y with networth_y
4	Disaggregated wealth	adds nla_y, eq_y, super_y, ha_y; drops aggregates
5	Spec 4 + full short-run dynamics	adds $\Delta^2 \log$ CCI, $\Delta \Delta_4$ income, $\Delta^2 \log$ unemp, $ \hat{\epsilon} $
6	Conventional constant-MPC disaggregated ECM (baseline)	Spec 5 + post-2008 PI break ln_yp_over_y_post2008; plain ha_y/eq_y/super_y/nla_y, CCI short-run only
6b	Spec 6 with back-extension-compatible SR CCI	replaces $\Delta^2 \log$ CCI with $\Delta^2 \log$ RBA D02 credit; disaggregated wealth proxies; fits on n = 180
7	Spec 6 + cohort terms + synthetic burden	adds prime_age_share, fhb_share
7b	Spec 7 with RBA E13 measured burden	post-2009 sample only
8	Williams CCI interactions (free)	plain real_rate/ln_hp_over_y replaced by $r \times \text{CCI}$ and $\log(\text{HP}/y) \times (1 - 1.2 \cdot \text{CCI})$, plus ha_x_cci and $\log(y^p/y) \times \text{CCI}$; plain ha_y retained
9	Spec 8 with Kalman state-space CCI	mirrors Spec 8's interaction set (incl. ha_x_cci_k) with cci_kalman in place of the spline
10	Williams-prior calibrated	$\gamma_{\text{IFA}} = 0.022$, $\psi_0 = 0.20$, $\psi_1 = 0.93$, $\omega = 1.2$; iterative fixed-point OLS
11	Faithful LIVES (free, headline)	housing via ha_x_cci only; ζ_c CCI intercept cci_williams restored; IFA combined (ilfa_y); nla_y, hp_x_1_minus_cci, r_x_cci, ln_yp_over_y, yp_x_cci
12	Williams-calibration-imposed LIVES	Spec 11 form with $\psi_0 = 0.20$, $\psi_1 = 0.93$, $\gamma_{\text{IFA}} = 0.022$ imposed via iterative fixed-point

The structural difference between Spec 6 and Spec 11 is decisive for the paper's central result. Spec 6 carries wealth as plain, constant marginal propensities and lets the credit-conditions index enter only as a short-run term. Under the LIVES theory the housing marginal propensity to consume is zero at $\text{CCI} = 0$ and is unlocked only as credit conditions ease, so housing enters

Spec 11 only via the interaction term ha_x_cci , the autonomous-consumption CCI loading ζ_c ($cci_williams$) is restored, and the two illiquid-financial components are combined into $ilfa_y$ as the theory specifies. Source: specification definitions; coefficient outputs in Table B4–B11 below.

1.2.2 B.2 The four selection screens

Each estimable specification is screened on four criteria, with BIC as tiebreak (source: specification-selection output):

1. **Sign screen** — every long-run coefficient with a non-ambiguous theoretical prior has the correct sign.
2. **Cointegration screen** — an Engle–Granger residual ADF test rejects the no-cointegration null at 5 per cent against MacKinnon (1991, 2010) critical values keyed to the regressor count. A Johansen trace statistic on one fixed trivariate subsystem (log consumption, log income, and either $ln_networth_y$ or ha_y ; $K = 2$, restricted constant) is reported alongside but tests only the $r = 0$ null of that small common subsystem, not each specification's own long run.
3. **Speed-of-adjustment screen** — λ has the correct (negative) sign and lies in $(0.02, 0.30)$.
4. **Stability screen** — Chow at 2008Q3 is not rejected at the 1 per cent level, and λ is sign-stable across at least 3 of the 4 sample variants (full, pre-COVID, COVID-dropped, COVID-rich-dummies). Where the standard `strucchange::sctest` design is singular, a manual common-coefficient Chow F-test is substituted (recorded in a `chow_method` field); a Chow that is incomputable even after the fallback is treated as neutral, not as a failure.

The cointegration battery covers Spec 8 and Spec 11 directly (both fail, like every other specification: Spec 11 EG-ADF -3.13 against a MacKinnon 5 per cent critical value of -5.47 for nine regressors; Spec 8 -3.40 against -5.70 for ten). Spec 9 is skipped (its de-meaned Kalman interactions are constructed locally and not available to the static long-run regression) as are Specs 10 and 12 (their long run is a calibrated offset, not a freely estimated static regression, so an Engle–Granger residual test is not applicable — see Table F3/F4 for the bounds-test alternative applied to Specs 6, 11 and 12). The speed-of-adjustment upper bound ($|\lambda| < 0.30$) binds for Specs 8 and 11 on the full sample ($|\lambda| = 0.458$ and 0.448); both are correctly signed, strongly significant and sign-stable across all four sample variants, and their pre-COVID λ ($-0.219, -0.266$) lies inside the screen interval.

1.2.3 B.3 Selector outcome

Table B3 — Four-screen outcome card. Source: specification- selection output; BIC is the Schwarz criterion.

Spec	Signs	Coint	λ	Stability	BIC
1	✓	×	✓	✓	-919.3
2	×	×	✓	✓	-501.7
3 (selector-preferred)	✓	×	✓	✓	-919.4
4	×	×	✓	✓	-905.3
5	×	×	✓	✓	-494.6
6 (conventional baseline)	×	×	✓	✓	-492.5
6b	✓	×	✓	×	-1114.0
7	×	×	×	✓	-500.6
7b	×	×	×	✓	-364.5
8	×	×	×	✓	-952.8
9	×	NA	✓	×	-890.6
10	✓	NA	✓	×	-493.2
11 (faithful LIVES, headline)	✓	×	×	×	-954.8
12	✓	NA	✓	×	-893.8

Under the canonical setting no specification passes all four screens; the selector falls back to a most-passes rule with BIC tiebreak. Spec 1 and Spec 3 each pass three screens, and Spec 3 (aggregate net worth in levels) is the automated pick (-919.4 against Spec 1's -919.3) — a conservative, non-LIVES form. Spec 6b carries the lowest raw BIC only because it is fitted on the longest ($n = 190$) back-extended sample and is not directly comparable to the $n = 146$ / $n = 86$ forms. Spec 11 (the faithful LIVES form) carries the best BIC of all fourteen specifications; what stands between it and the automated pick is the conservative $|\lambda|$ ceiling (breached by the COVID-inflated full-sample estimate) and the 2008Q3 Chow test. The manuscript leads with Spec 11 on theoretical-form grounds, retains Spec 6 as the conventional baseline, and carries Spec 12 (and its independent reproduction, Spec 10) as the negative control.

1.2.4 B.4 Headline results — Spec 11 (faithful LIVES)

Table B4 — Spec 11 full-sample long-run coefficients. Full sample 1988Q3–2024Q4, $n = 146$, $\text{adj-}R^2 = 0.824$. Structural $\gamma = \text{OLS}/|\lambda|$.

Term	OLS coef	t-stat	Implied γ	Williams
ha_x_cci (γ_1 , housing $\times$ CCI)	+0.0025	+0.71	0.005	0.049
nla_y (γ_{NLA})	+0.0269	+3.75	0.060	0.159
ilfa_y (γ_{IFA})	+0.0155	+3.09	0.035	0.022
cci_williams (ζ_c)	+0.0001	+0.01	0.000	0.190
hp_x_1_minus_cci (α_4)	+0.0279	+3.08	0.062	-0.130
r_x_cci (α_1)	+0.0028	+4.55	0.006	-0.871
ln_yp_over_y (ψ_0)	+0.459	+4.04	1.024	0.20
yp_x_cci (ψ_1)	-0.510	-1.51	-1.138	0.93
ecm_lag (λ)	-0.448	-3.57	(= 1)	-0.286

Table B5 — Spec 11 across the four sample treatments. Newey–West HAC t -statistics in parentheses.

Variant	n	λ (t)	nla_y (t)	ilfa_y (t)	ha_x_cci ln_yp_over_y (t)	(t)
Full sample	146	-0.448 (-3.6)	+0.0269 (3.8)	+0.0155 (3.1)	+0.0025 (0.7)	+0.459 (4.0)
Pre-COVID (to 2019Q4)	126	-0.266 (-4.8)	+0.0159 (1.8)	+0.0093 (1.7)	+0.0019 (0.9)	+0.298 (5.8)
COVID quarters dropped	138	-0.248 (-6.7)	+0.0170 (2.1)	+0.0098 (2.0)	+0.0023 (1.1)	+0.281 (8.7)
Quarterly COVID dummies	146	-0.242 (-6.2)	+0.0141 (3.1)	+0.0084 (3.3)	+0.0016 (0.8)	+0.278 (8.2)

The full-sample λ (-0.448) is inflated by the COVID quarters — the three pulse dummies are demonstrably insufficient, and replacing them with quarterly dummies halves $|\lambda|$. The identified speed of adjustment is $\lambda \approx -0.25$ (pre-COVID/COVID-dropped/COVID-rich cluster, t -ratios -4.8 to -6.7), within roughly 7 per cent of Williams' -0.286; the net-liquid and illiquid-financial m.p.c.s are correctly signed in every variant and significant at 5 per cent in three of the four. Three qualifiers: (i) the housing-collateral interaction ha_x_cci is correctly signed but insignificant in every variant (implied peak m.p.c. 0.0055 against Williams' 0.049); the real-rate interaction is wrong-signed and significant in every variant; the affordability interaction is wrong-signed on the full sample; yp_x_cci is wrong-signed full-sample but flips to the theory-correct sign pre-COVID and COVID-dropped. (ii) The permanent-income gearing $\hat{\psi} = \text{OLS}/|\lambda|$ is 1.02 (full) to 1.13 (COVID-controlled) — above the admissibility bound $\psi \leq 1 - \eta \approx 0.95$ (§E, §F). (iii) The wealth-magnitude comparison with Williams is governed by the nested intervals of Table F7, under which neither magnitude can be distinguished from Williams' values (the fixed-regressor comparison of Table F2 is superseded).

1.2.5 B.5 The calibration route (Spec 12, Spec 10)

Spec 12 imposes Williams' scale-robust calibrations ($\gamma_{\text{IFA}} = 0.022$, $\psi_0 = 0.20$, $\psi_1 = 0.93$) via an iterative fixed-point offset and frees only the housing-collateral m.p.c., the net-liquid m.p.c. and λ . His real-rate, affordability and intercept loadings cannot be imposed at their published magnitudes without diverging the fixed point (the percent-scaled real rate against a unit-normalised CCI makes his $\alpha_1 = -0.871$ roughly thirty times too large); the CCI-support problem compounds this — on the deployed index (range $[-2.1, 1]$ versus Williams' $[0, 0.8]$) the imposed gearing $\psi(\text{CCI}) = 0.20 + 0.93 \cdot \text{CCI}$ is negative in the post-2019 regime and breaches Williams' own $\psi \leq 0.95$ cap at the peak.

Result: imposing Williams' permanent-income calibration collapses the error-correction to $\lambda \approx 0$ (Spec 12: $\lambda = -0.030$, $t = -0.74$, flipping sign pre-COVID; independently reproduced by Spec 10, $\lambda = -0.048$, $t = -0.78$, which keeps the rate and affordability channels free). The Australian data freely estimate a structural permanent-income gearing of order one — several times Williams' 0.20 — so forcing his value injects a large, mis-signed contribution that destroys the equilibrium. The LIVES *structure* transfers; Williams' Australian *calibrations* do not.

1.2.6 B.6 Baseline results — Spec 6 (conventional)

Table B6 — Spec 6 long-run coefficients. Full sample 1988Q3–2024Q4, $n = 86$ (binding constraint: `cci_ratio` from ABS Cat 5601.0, available from 2002Q3).

Term	OLS coef	NW SE	t-stat	Implied γ	Sign OK
ha_y	+0.0022	0.0076	+0.30	+0.009	✓
nla_y	+0.0083	0.0351	+0.24	+0.035	✓
eq_y	−0.0156	0.0517	−0.30	−0.065	✗
super_y	+0.0060	0.0091	+0.66	+0.025	✓
ln_hp_over_y	+0.0102	0.0444	+0.23	+0.043	n/a
real_rate	−0.00053	0.0011	−0.46	−0.0022	✓
ln_yp_over_y	+0.3253	0.2173	+1.50	+1.363	n/a
ln_yp_over_y_post2008	+0.1704	0.1965	+0.87	+0.714	n/a
ecm_lag (λ)	−0.2386	0.0935	−2.55	(=1)	✓

(Short-run regressors and event dummies omitted; full vector in Table B10.)

Speed of adjustment: $\lambda = -0.239$ ($t = -2.55$, $p = 0.013$), about 83 per cent of Williams' -0.286 , but significance leans on the COVID quarters — the pre-COVID estimate collapses to -0.087 ($t = -0.79$), against Spec 11's pre-COVID -0.266 ($t = -4.85$). Housing wealth: OLS $+0.0022$ ($t = 0.30$), statistically indistinguishable from zero as a standalone level — the coefficient the theory predicts to be ≈ 0 absent the CCI interaction. Net liquid assets: OLS $+0.0083$ ($t = 0.24$), implied $\gamma_{\text{NLA}} = 0.035$, about a fifth of Williams' 0.159, correctly signed; the $\gamma_{\text{LA}} + \gamma_{\text{LOANS}} = 0$ restriction is accepted at 5 per cent (Table C5). Illiquid financial wealth decomposes into equities ($\gamma = -0.065$, wrong sign, $t = -0.30$) and superannuation ($\gamma = +0.025$, $t = 0.66$); the combined $\gamma_{\text{IFA}} = -0.040$ is wrong-signed, dragged below zero by the equities component —

a small-sample artefact of the disaggregated split that Spec 11's combined `ilfa_y` removes (+0.0155, $t = 3.09$). Permanent income: base coefficient +0.325 ($t = 1.50$) plus a post-2008 break of +0.170 ($t = 0.87$), neither individually significant; implied structural gearing at CCI = 0 is 1.36, well above Williams' calibrated 0.20. λ is sign-stable across all four sample variants (full -0.239, pre-COVID -0.087, COVID-dropped -0.162, COVID-rich -0.177).

1.2.7 B.7 Diagnostics summary — all fourteen specifications

Table B7 — Diagnostics, full sample. AR(1)/AR(4) are Breusch–Godfrey p -values; "Het p " is the Breusch–Pagan-type LM p -value with the `het_diagnosis` field in parentheses. A `chow_method` field (not shown) records whether the 2008Q3 stability statistic used `strucchange::sctest` or a manual split-sample Chow F-test where the standard design was singular.

Spec	n	adj R ²	DW	AR(1) p	AR(4) p	RESET p	Het p (diagnosis)	BIC
1 (LogNet-Worth)	146	0.731	2.32	0.026	0.093	<0.001	<0.001 (struct)	-919.3
2 (LogNetWorth_CCI)	86	0.772	2.44	0.020	0.165	0.003	<0.001 (struct)	-501.7
3 (Level-NetWorth)	146	0.731	2.32	0.025	0.088	<0.001	<0.001 (struct)	-919.4
4 (Disagg_NoCCI)	146	0.726	2.32	0.018	0.067	<0.001	<0.001 (struct)	-905.3
5 (FullDisagg)	86	0.802	2.25	0.107	0.087	<0.001	<0.001 (struct)	-494.6
6 (conventional baseline)	86	0.804	2.15	0.339	0.301	<0.001	<0.001 (struct)	-492.5
6b (LongHist-SRCCI)	180	0.703	2.12	0.275	0.134	<0.001	<0.001 (struct)	-1114.0
7 (Cohort-Burden)	86	0.834	2.21	0.203	0.050	0.018	0.002 (struct)	-500.6
7b (RBABurden)	64	0.872	2.16	0.327	0.004	0.109	0.003 (struct)	-364.5
8 (CCI_Interactions)	146	0.827	1.92	0.494	0.057	0.001	<0.001 (struct)	-952.8
9 (KalmanCCI)	146	0.735	2.18	0.140	0.018	<0.001	<0.001 (struct)	-890.6
10 (WilliamsPrior)	86	0.773	2.22	0.186	0.142	0.035	<0.001 (struct)	-493.3
11 (LIVES_Headline)	146	0.824	1.90	0.448	0.055	0.001	<0.001 (struct)	-954.8
12 (LIVES_Calibrated)	146	0.687	2.14	0.317	0.010	0.017	<0.001 (struct)	-893.8

Heteroscedasticity is structural in every full-sample specification — the LM rejection survives dropping the event quarters in every case — so Newey–West HAC standard errors are used throughout. RESET rejects in every specification bar the short-sample Spec 7b ($p = 0.109$); the faithful Spec 11 is not exempt ($p = 0.001$). Low-order serial correlation (Breusch–Godfrey AR(1)) is concentrated in the aggregated and disaggregated-no-CCI forms (Specs 1–4); at fourth order the rejections sit instead in Specs 7, 7b, 9, 12, with Specs 8 and 11 borderline

($p = 0.057, 0.055$). The headline LIVES form (Spec 11) carries the best BIC of any $n = 146$ specification, including the over-parameterised Spec 8 (-954.8 vs -952.8).

1.2.8 B.8 Comparison with Williams (2010, 2012) – conventional baseline

Williams reports structural long-run coefficients γ ; the OLS coefficients relate to those γ by the ECM identity $\text{OLS_coef} = \lambda \times \gamma$, so reporting both forms separates the two channels.

Table B8 – Spec 6 vs Williams' Table 1.

Term	Williams γ	Williams implied OLS	Our OLS	Our γ	OLS gap	γ gap
λ	-0.2860	(same)	-0.2386	(same)	-17 %	(same)
Housing wealth ha_y	0.0488	0.0140	0.0022	0.0094	-84 %	-81 %
Illiquid eq_y + super_y	0.0220	0.0063	-0.0096	-0.0403	wrong sign	wrong sign
Net liquid nla_y	0.1590	0.0455	0.0083	0.0350	-82 %	-78 %
log(HP/y)	-0.1300	-0.0372	+0.0102	+0.0429	wrong sign	wrong sign
ψ at CCI = 0	0.2000	0.0572	0.3253	1.3632	—	—

Williams' $\gamma_{\text{HA}} = 0.0488$ is the paper's derived peak housing MPC, not the raw Table 1 coefficient ($\gamma_1 = 0.0606$); γ_{NLA} and γ_{IFA} are Table 1's raw coefficients. Williams' Table 1 appears in the full working- paper version of Muellbauer and Williams (2012) (CEPR Discussion Paper 8386, revised 12 April 2012); the seven-page *BIS Papers* No. 64 chapter is a condensed version that does not itself contain Table 1.

The speed of adjustment is the coefficient that agrees with Williams (-0.239 sits 17 per cent below his -0.286 , well inside one standard error); the disaggregated wealth OLS coefficients sit roughly 80 per cent below his implied OLS values, and because the λ gap is small the ECM identity does not rescale the wealth deficits away. Two readings discipline this: none of the wealth gaps is sharply estimated (the Spec 6 delta-method intervals in Table F1 are wide enough to contain Williams' value and zero for every wealth coefficient), and Spec 6 is not itself the LIVES equation – its housing channel is a standalone level the theory predicts to be ≈ 0 , so neither a match nor a miss carries structural content. Spec 11 (Table B4, Table F2) is the comparison that carries structural weight.

1.2.9 B.9 λ across all fourteen specifications

Table B9 – Speed of adjustment λ (ecm_lag), full and pre-COVID samples.

Specification	λ full (n=146)	λ pre-COVID (n=126)
Spec 1	-0.193 (t=-2.88)***	-0.161 (t=-3.32)***
Spec 2	-0.186 (t=-2.07)**	-0.080 (t=-1.15)
Spec 3	-0.191 (t=-2.90)***	-0.158 (t=-3.17)***
Spec 4	-0.182 (t=-2.45)**	-0.119 (t=-1.95)*
Spec 5	-0.235 (t=-2.47)**	-0.061 (t=-0.62)
Spec 6	-0.239 (t=-2.55)**	-0.087 (t=-0.79)
Spec 6b	-0.248 (t=-3.96)***	-0.240 (t=-4.61)***
Spec 7	-0.341 (t=-2.76)***	-0.054 (t=-0.43)
Spec 7b	-0.381 (t=-2.43)**	-0.061 (t=-0.93)
Spec 8	-0.458 (t=-3.52)***	-0.219 (t=-3.95)***
Spec 9	-0.209 (t=-3.24)***	-0.158 (t=-2.86)***
Spec 10	-0.048 (t=-0.78)	-0.025 (t=-0.57)
Spec 11	-0.448 (t=-3.57)***	-0.266 (t=-4.85)***
Spec 12	-0.030 (t=-0.74)	+0.041 (t=+2.03)**

*** sig 1%, ** sig 5%, * sig 10%. Several specifications estimate on different windows from the nominal $n = 146/n = 126$: Specs 2, 5, 6, 7 and 10 on $n = 86$ because a 2002Q3-binding credit term shortens their sample, Spec 7b on $n = 64$, and Spec 6b on the $n = 180$ back-extended sample.

1.2.10 B.10 Long-run coefficient matrix — five disaggregated specifications

Table B10 — Long-run coefficients, full sample. "—" = term not in that specification. *** sig 1%, ** sig 5%, * sig 10%.

Term	Spec 4	Spec 5	Spec 6	Spec 8	Spec 11
nla_y	+0.0045 (+0.39)	−0.0141 (−0.43)	+0.0083 (+0.24)	+0.0337 (+4.39)***	+0.0269 (+3.75)***
eq_y	−0.0079 (−0.41)	−0.0295 (−0.57)	−0.0156 (−0.30)	−0.0014 (−0.11)	—
super_y	+0.0025 (+0.36)	+0.0040 (+0.37)	+0.0060 (+0.66)	+0.0215 (+2.42)**	—
ilfa_y	—	—	—	—	+0.0155 (+3.09)***
ha_y	+0.0035 (+0.53)	−0.0028 (−0.30)	+0.0022 (+0.30)	+0.0030 (+0.48)	—
ha_x_cci	—	—	—	+0.0043 (+0.87)	+0.0025 (+0.71)
ln_hp_over_y	+0.0038 (+0.13)	+0.0447 (+0.83)	+0.0102 (+0.23)	—	—
hp_x_l_minus_cci	—	—	—	+0.0299 (+1.38)	+0.0279 (+3.08)***
real_rate	−0.0002 (−0.33)	−0.0014 (−0.87)	−0.0005 (−0.46)	—	—
r_x_cci	—	—	—	+0.0028 (+2.05)**	+0.0028 (+4.55)***
cci_williams	—	—	—	—	+0.0001 (+0.01)
ln_yp_over_y	+0.2026 (+1.94)*	+0.3316 (+1.43)	+0.3253 (+1.50)	+0.4921 (+3.42)***	+0.4591 (+4.04)***
ln_yp_over_y_post2008	—	—	+0.1704 (+0.87)	—	—
yp_x_cci	—	—	—	−0.5046 (−2.43)**	−0.5101 (−1.51)
ecm_lag	−0.1824 (−2.45)**	−0.2353 (−2.47)**	−0.2386 (−2.55)**	−0.4583 (−3.52)***	−0.4483 (−3.57)***

1.2.11 B.11 Spec 11 vs Spec 12 — selected coefficients

Table B11 — Faithful LIVES (Spec 11) vs Williams-calibration-imposed (Spec 12), full and pre-COVID samples. Convention: $\lambda = \text{ecm_lag}$ (negative = error-correction); structural $\gamma = \text{OLS}/|\lambda|$. *** sig 1%, ** sig 5%, * sig 10%.

Term	Spec 11 full (n=146)	Spec 11 pre- COVID (n=126)	Spec 12 full (n=146)	Spec 12 pre- COVID (n=126)
λ (ecm_lag)	-0.448 (t=-3.57)***	-0.266 (t=-4.85)***	-0.030 (t=-0.74)	+0.041 (t=2.03)**, sign flip
nla_y	+0.0269 (t=3.75)***	+0.0159 (t=1.81)*	+0.0011 (t=0.39)	-0.0002 (t=-0.09, wrong sign)
ilfa_y (=eq+super)	+0.0155 (t=3.09)***	+0.0093 (t=1.74)*	imposed ($\gamma=0.022$)	imposed ($\gamma=0.022$)
ha_x_cci (γ_1)	+0.0025 (t=0.71)	+0.0019 (t=0.86)	-0.0012 (t=-0.88)	+0.0018 (t=1.52)
hp_x_l_minus_cci	+0.0279 (t=3.08)***	+0.0137 (t=1.10)	—	—
r_x_cci	+0.0028 (t=4.55)***	+0.0022 (t=3.47)***	—	—
cci_williams (ζ_c)	+0.0001 (t=0.01)	+0.0199 (t=2.77)***	—	—
ln_yp_over_y	+0.4591 (t=4.04)***	+0.2982 (t=5.81)***	imposed ($\psi_0=0.20$)	imposed ($\psi_0=0.20$)
yp_x_cci	-0.5101 (t=-1.51)	+0.1869 (t=1.80)*	imposed ($\psi_1=0.93$)	imposed ($\psi_1=0.93$)
Intercept	-0.0139 (t=-2.13)**	-0.0099 (t=-1.54)	+0.0078 (t=1.81)*	+0.0048 (t=2.13)**
adj-R ²	0.824	0.239	0.687	0.073
SE (%)	0.683	0.510	0.911	0.565
BIC	-954.75	-900.15	-893.79	-896.61

The contrast is the structure-transfers-but-calibrations-do-not result: imposing Williams' Australian calibrations via the iterative fixed point collapses the error-correction speed from -0.448 to a statistically insignificant -0.030, and flips it to the wrong sign and significant +0.041 ($t = 2.03$) pre-COVID.

1.3 C. Robustness battery

The robustness suite runs on the automated-selector specification (Spec 3), on the conventional baseline (Spec 6) where the battery requires the disaggregated wealth split, and — in parallel — on the faithful LIVES specification (Spec 11).

1.3.1 C.1 OLS vs IV on current income

Because current income enters both the error-correction term and the permanent-income gap, both regressors are instrumented (income lagged one, two and four quarters, unemployment

lagged one and two quarters, mortgage rate lagged one quarter — six instruments; both OLS and IV columns carry Newey–West HAC standard errors).

Frame	λ (OLS)		λ (IV)	Change	$\ln_yp_over_y$ (OLS)	$\ln_yp_over_y$ (IV)
Spec 3, n=146	-0.191	-0.258	(SE 0.110)	+35%	+0.196	+0.280 (+43%)
Spec 11, n=146	-0.448		-0.503	+12%	—	—

Spec 3 diagnostics: weak-instrument $F = 73.6$ ($\ln_yp_over_y$), 28.9 (ecm_lag); Wu–Hausman does not reject OLS exogeneity at 5 per cent ($p = 0.095$); Sargan does not reject instrument validity at 5 per cent ($p = 0.074$). Spec 11 diagnostics (endogenous set: ecm_lag , $\ln_yp_over_y$, yp_x_cci , ha_x_cci , $hp_x_1_minus_cci$; one overidentifying restriction): Wu–Hausman rejects exogeneity ($p = 0.002$); weak-instrument $F = 126.0$ ($\ln_yp_over_y$), 55.5 (ha_x_cci), 32.5 (ecm_lag), 15.2 ($hp_x_1_minus_cci$), but only 6.5 for yp_x_cci (below the conventional weak-instrument threshold); nla_y attenuates to insignificance ($0.0269 \rightarrow 0.0040$, $t \approx 0.3$); ha_x_cci and $hp_x_1_minus_cci$ flip sign relative to OLS ($+0.0025 \rightarrow -0.013$; $+0.028 \rightarrow -0.015$); Sargan $p = 0.251$ (near-vacuous at $df = 1$). The Spec 11 IV design is underpowered: current-income endogeneity is a live concern (Wu–Hausman), but the wealth-channel point estimates cannot bear weight under instrumentation.

1.3.2 C.2 Joint permanent-income + consumption SUR

Estimating the consumption equation jointly with a one-step-ahead income-growth forecasting equation by SUR leaves every coefficient within sampling noise of single-equation OLS.

Frame	λ (OLS)	λ (SUR)	$\Delta\%$	$\ln_yp_over_y$ $\Delta\%$
Spec 3, n=146	-0.191	-0.204	+7.0%	-4.5%
Spec 11, n=146	-0.448	-0.457	+1.9%	-1.4% (nla_y -0.2%, $ilfa_y$ +2.6%)

The second equation is a one-step-ahead forecast, not the $k = 40$ - quarter discounted projection that defines the permanent-income regressor, so the SUR tests residual covariance, not generated- regressor bias (the Table F1/F2 caveat applies). Single-equation OLS is an acceptable estimator for the consumption block; the case for the multi-equation framework rests on cross-equation parameter restrictions, not on residual covariance or efficiency gain (§D.4, §C.13).

1.3.3 C.3 Chow battery and multiple-break tests

Chow tests on Spec 3 do not reject stability at 1995Q1 (stat 0.380, $p = 0.978$), 2000Q1 (0.740, $p = 0.730$) or 2008Q3 (0.529, $p = 0.912$), while 2020Q1 is strongly rejected (11.12, $p = 8.9 \times 10^{-16}$). On Spec 11, 1995Q1 ($p = 0.844$) and 2000Q1 ($p = 0.152$) are stable, but 2008Q3 is rejected at 5 per cent (1.914, $p = 0.017$) — consistent with the GFC being where Spec 11's credit-interaction structure starts to bind (the deployed CCI's first knot is 2007Q3, §D.2) — and 2020Q1 is not computable on the COVID-dummied subsample. The Bai–Perron/CUSUM battery on Spec 3 finds a single dominant break: $\sup F = 169.96$ ($p = 0$) dated 2019Q1 (the trimming convention places the date at the segment boundary adjacent to COVID), with CUSUM $p = 0.971$ (recursive residuals stable away from that episode).

1.3.4 C.4 Williams CCI interactions (Spec 8) — reallocation, not identification

Table C1 — Sign-prior verdicts, Spec 8 free interactions.

Williams interaction	Sign prior	OLS coef	t	p	Verdict
$r \times \text{CCI}$	–	+0.00283	+2.05	0.042	wrong sign, significant – FAIL
$\log(\text{HP}/y) \times (1 - 1.2 \cdot \text{CCI})$	–	+0.0299	+1.38	0.171	wrong sign, insignificant – FAIL
$\log(y^p/y) \times \text{CCI}$	+	–0.5046	–2.43	0.016	wrong sign, significant – FAIL
$\log(y^p/y)$	+	+0.4921	+3.42	0.001	right sign – PASS
$\text{HA} \times \text{CCI} (\gamma_i)$	+	+0.00426	+0.87	0.384	right sign, insignificant

Three of the four sign-priced interactions fail. Against Williams' Table 1 the raw-coefficient gaps remain structural: $\alpha_{c1} (r \times \text{CCI}) -0.871$ vs our structural $+0.0062$; $\alpha_{c4} (\text{HP}/y \times (1 - 1.2 \cdot \text{CCI})) -0.13$ vs $+0.065$; $\psi_1 +0.93$ (calibrated) vs -1.101 (freely estimated). Spec 8 re-allocates the long-run identification rather than recovering Williams' channels: standalone nla_y strengthens (struct $+0.073$), $super_y$ strengthens (struct $+0.047$), and λ shifts from -0.239 (Spec 6) to -0.458 – past Williams' magnitude – while attaining the highest adj- R^2 (0.827) among the $n = 146$ specifications. Pre-COVID, the standalone housing level becomes significant ($ha_y +0.0154^{***}$, $t = 2.87$) but the interactions turn negative ($ha_x_cci -0.0069$, $t = -1.42$; $hp_x_1_minus_cci -0.0292$, $t = -1.54$) while λ halves to -0.219 ($t = -3.95$). The six CCI-interacted regressors act as flexible parameter time-variation, not as a structurally identified common-factor channel (§D.5).

1.3.5 C.5 Net-liquid-assets restriction $\gamma_{LA} + \gamma_{LOANS} = 0$

Table C2 — Wald test of $H_0: \gamma_{LA} + \gamma_{LOANS} = 0$ (car::linearHypothesis, Newey–West variance).

Spec	Sample	$\gamma_{LA} + \gamma_{LOANS}$	NW SE	t	p	Restriction
4	full	+0.0325	0.0445	0.731	0.465	accepted
5	full	+0.0422	0.0999	0.422	0.673	accepted
6	full	+0.0202	0.0987	0.204	0.838	accepted
4	pre-COVID	+0.0282	0.0285	0.989	0.322	accepted
5	pre-COVID	–0.0222	0.0499	–0.445	0.656	accepted
6	pre-COVID	–0.0026	0.0509	–0.051	0.959	accepted

Accepted at the 5 per cent level in every specification $\times$ sample combination – non-rejection-by-imprecision, but it validates the Italian convention of netting deposits against debt and supports the constructed nla_y series.

1.3.6 C.6 Drehmann amortising-mortgage adjusted real rate

$N = 25$ years (100 quarters), Spec 3, $n = 146$. The substitution is essentially inert: ecm_lag -0.1906 (base) $\rightarrow -0.1902$ (Drehmann); net-worth $0.00108 \rightarrow 0.00129$; $ln_yp_over_y$ $0.1963 \rightarrow 0.1959$; real-rate $-0.00020 \rightarrow -0.00015$ (insignificant either way). Undefined for the faithful LIVES specification, which carries no plain $real_rate$ term (its rate enters only through $r \times CCI$).

1.3.7 C.7 Scaled-income robustness

Averaging disposable income with labour-plus-transfer income (Spec 3, $n = 146$) shifts λ from -0.1906 to -0.2111 and $ln_yp_over_y$ from 0.196 to 0.163 , with net-worth roughly halving ($0.00108 \rightarrow 0.00061$). On Spec 11, λ moves from -0.448 to -0.525 while the wealth m.p.c.s strengthen (nla_y $0.0269 \rightarrow 0.0344$, $ilfa_y$ $0.0155 \rightarrow 0.0175$) and $ln_yp_over_y$ eases ($0.459 \rightarrow 0.376$). The income-measure choice moves $|\lambda|$ by 0.02 – 0.08 in level but does not change the substantive ranking or signs of the wealth coefficients in either frame.

1.3.8 C.8 Williams non-property income (NPY) robustness

Replacing disposable income with np_real_pc (Williams 2009 §4.2.1) is close to inert on Spec 3's speed of adjustment ($-0.1906 \rightarrow -0.1857$, -2.6%), with $ln_yp_over_y$ easing ($0.196 \rightarrow 0.164$) and net-worth staying negligible ($0.00108 \rightarrow 0.00044$). On Spec 11 the NPY measure trims λ ($-0.448 \rightarrow -0.395$) while raising the wealth m.p.c.s (nla_y $0.0269 \rightarrow 0.0413$, $ilfa_y$ $0.0155 \rightarrow 0.0203$) and easing $ln_yp_over_y$ ($0.459 \rightarrow 0.352$). The income concept is not what separates the paper's estimates from Williams': under his own income measure the conventional λ is essentially unchanged and the LIVES wealth channels strengthen.

1.3.9 C.9 Permanent-income method comparison

See §E, Table E1.

1.3.10 C.10 Permanent-income filter sensitivity

A grid over discount factor $\delta \in \{0.90, 0.95, 0.97\}$ and horizon $k \in \{20, 40, 60\}$ quarters, ogive on/off, under the AR constructor on Spec 3: λ ranges only from -0.0470 to -0.0482 across the eighteen cells (baseline $\delta = 0.95$, $k = 40$, ogive on: $\lambda = -0.0479$, structural PI weight -0.076 , structural net-worth weight $+0.122$). The ogive toggle moves λ by less than 0.0003 and the structural PI weight by at most 0.03 within each $\delta \times k$ cell. The PI weight is more sensitive to the horizon (-0.21 at $k = 20$ down to -0.04 at $k = 60$) but never changes sign within the AR method. An HP-filter ($\lambda = 1600$) permanent income moves λ to -0.0820 and flips the structural PI weight to $+1.238$. The within-AR-method PI tuning is not what drives the $|\lambda|$ gap with Williams; the dominant factor is the AR-versus-Italy-direct-forecast method choice (§E).

For the headline LIVES specification: removing the GFC ogive moves Spec 11's λ from -0.448 to -0.574 ($t = -4.63$) and $ln_yp_over_y$ from $+0.459$ to $+0.604$, leaving the structural gearing essentially unchanged ($\hat{\psi} \approx 1.05$ against the headline 1.02) – the $\$B.4/\E admissibility-bound breach is not an ogive artefact.

1.3.11 C.11 COVID-period robustness

All fourteen specifications carry all four sample variants (full, pre-COVID, COVID-dropped, COVID-rich). λ is sign-stable across all four for every specification except the two calibration-imposed ones. Spec 6: $-0.239 / -0.087 / -0.162 / -0.177$ (all correctly signed, though pre-COVID

is small and insignificant). Spec 11: $-0.448 / -0.266 / -0.248 / -0.242$ (sign-stable, full-sample inflated, the tightly clustered -0.24 to -0.27 of the three COVID-controlled variants treated as the identified value). Spec 10 (sign-flips to $+0.009$ COVID-dropped, $+0.051$ COVID-rich) and Spec 12 (sign-flips to $+0.041$ pre-COVID, $+0.015$ COVID-rich) are not sign-stable — an artefact of the calibration collapse (§B.5), not of the COVID episode itself.

1.3.12 C.12 Rolling-window estimation

A 60-quarter rolling estimation of Spec 3 (87 windows ending 2003Q2–2024Q4) shows λ holding in a band of roughly -0.13 to -0.27 for windows ending before 2020; the net-worth coefficient trends down from about $+0.012$ in the earliest windows to near zero through the macroprudential era and mildly negative in the latest windows (-0.011 at 2024Q4, within one SE of zero). The COVID quarters destabilise the short windows: λ briefly flips to $+0.15$ around 2021Q3 before settling at -0.44 (SE 0.19) in the final window, where COVID dominates a 60-quarter span — read as a symptom of limited identifying variation in the post-deregulation sample, not model instability.

1.3.13 C.13 Out-of-sample forecast validation

Table C3 — Rolling out-of-sample RMSE, 36 expanding-window cuts ($h \in \{1, 4, 8\}$; $n = 36$ at $h = 1, 4$; $n = 32$ at $h = 8$).

Specification	$h = 1$ RMSE	$h = 4$ RMSE	$h = 8$ RMSE
Benchmark RW drift	0.03094	0.03094	0.03282
Benchmark AR(1)	0.03703	0.03102	0.03283
Spec 4 (disagg, no CCI)	0.03175	0.03182	0.03896
Spec 6 (conventional baseline)	0.03231	0.03293	0.04180
Spec 7 (cohort-burden)	0.03247	0.03164	0.03540
Spec 8 (Williams CCI)	0.02901	0.03323	0.04038
Spec 9 (Kalman CCI)	0.03206	0.03299	0.03929
Spec 11 (LIVES headline)	0.02919	0.03517	0.06402

At $h = 1$, Spec 8 (0.0290) and Spec 11 (0.0292) beat the RW-drift benchmark (0.0309). At $h = 4$ and $h = 8$ the random walk with drift dominates every structural specification (best structural at $h = 4$: Spec 7, 0.0316 vs 0.0309; at $h = 8$: Spec 7, 0.0354 vs 0.0328), and Spec 11 is the worst performer at $h = 8$ (0.0640). A construction caveat applies to the whole exercise: the validator's permanent-income input is the full-sample Italy-method measure, not a real-time forecaster, and the credit-conditions series and its de-mean constants are likewise full-sample objects. Both generated regressors therefore embed information from beyond each expanding-window cut, so these RMSEs measure fit stability under re-estimation, not genuine real-time forecast accuracy — an upper bound on what a fully real-time forecaster would deliver.

1.3.14 C.14 Back-extension robustness — Spec 1 on the 1976Q3+ sample

Table C4 — Spec 1 (aggregate net worth), structural $\gamma = \text{OLS}/|\lambda|$.

LR coefficient (γ)	1988+ baseline (n=146)	1976+ extended (n=190)	% change
λ (ecm_lag)	-0.1934	-0.2090	+8.1
ln_networth_y	+0.0195	+0.0467	+140
ln_hp_over_y	+0.0443	+0.0375	-15.5
real_rate	-0.00115	+0.00005	sign flip
ln_yp_over_y	+1.0379	+0.9913	-4.5

The speed of adjustment and permanent-income elasticity are essentially stable across samples (λ -0.193 $\rightarrow$ -0.209; γ_{yp} 1.04 $\rightarrow$ 0.99); the aggregate wealth elasticity more than doubles (0.019 $\rightarrow$ 0.047) but both values are small and imprecisely estimated (baseline $t = 0.17$ on the OLS coefficient) – read as the pre-deregulation regime adding identifying variation to a coefficient the modern sample barely pins down, not as datable parameter instability.

1.3.15 C.15 Spec 4 on the back-extended sample

Structural γ : λ moves 11.5 per cent toward Williams (-0.1824 $\rightarrow$ -0.2034, still 29 per cent short of -0.286), but the individual wealth coefficients become smaller – γ_{NLA} collapses 95 per cent (+0.0245 $\rightarrow$ +0.0013), γ_{SUPER} flips sign (+0.014 $\rightarrow$ -0.005), γ_{HA} eases (+0.019 $\rightarrow$ +0.018), γ_{EQ} retains its wrong sign (-0.043 $\rightarrow$ -0.041). Sample length is not the binding constraint on whether the disaggregated single-equation form reproduces Williams' Table 1: the longer sample sharpens the speed of adjustment but blunts, rather than sharpens, the individual wealth channels.

1.3.16 C.16 Spec 6b – conventional baseline on the back-extended sample

Table C5 – Spec 6 vs Spec 6b, structural γ . $n = 180$ full / $n = 160$ pre-COVID for Spec 6b.

LR coefficient (γ)	Spec 6 (n=86)	Spec 6b (n=180)	Williams Table 1
λ (ecm_lag)	-0.239 ($t=-2.55$)	-0.248 ($t=-3.96$)	-0.286
ha_y / ha_y_proxy γ	0.009	0.012	0.049
nla_y / nla_y_proxy γ	0.035	0.015	0.159
eq_y / eq_y_proxy γ	-0.065	-0.008	(calibrated 0.011)
super_y / super_y_proxy γ	0.025	-0.001	(calibrated 0.011)
ln_hp_over_y γ	+0.043	+0.024	-0.130
ln_yp_over_y (CCI=0) γ	+1.363	+1.113	+0.20 (calibrated)
BIC	-492.5	-1114.0	n/a

The speed of adjustment moves modestly closer to Williams (87 per cent of -0.286 vs 83 per cent on Spec 6) and is far more sharply estimated (t improves from -2.55 to -3.96), identified in every sample variant (pre-COVID -0.240, $t = -4.61$; COVID-dropped -0.240; COVID-rich -0.234) – unlike Spec 6, whose λ collapses without the COVID quarters. The wealth γ profile shifts toward still-smaller individual elasticities (γ_{NLA} falls from 0.035 to 0.015, γ_{SUPER} flips sign) while γ_{HA} stays positive but at a quarter of Williams' 0.049. eq_y_proxy is held constant at its 1988Q3 value pre-1988 (§A.7), so the near-zero back-extended γ_{EQ}

and γ_{SUPER} should be read as proxy-limited rather than as estimates of a true pre-1988 propensity.

1.3.17 C.17 Maximal-GETS placebo on the back-extended sample

Table C6 — Extended-sample placebo (1976Q3+).

Placebo variant (1976Q3+)	Canonical adj-R ²	adj-R ² percentile	$ \lambda $ percentile	Verdict	
Extended (Williams literal 4-knot)	0.6801	36th	26th	below median on both	
Maximal-GETS (canonical 15- knot reduction)	0.6836	48th	70th	below R ² me- dian, above	

On the back-extended sample the literal Williams 4-knot construction sits below the placebo median on both metrics (adj-R² 0.6801 vs placebo median 0.6815; $|\lambda|$ 0.2023 vs 0.2079). The maximal-GETS reduction does better on mean reversion ($|\lambda|$ 0.2563 beats 70 per cent of random draws against a placebo median of 0.2329) but still sits just below the random median on fit (48th percentile), while retaining fewer knots (7) than the placebo median (8). Neither extended-sample construction is the deployed CCI: the deployed-protocol placebo on the 1988Q3+ estimation sample reaches the 84th adjusted-R² and 80th $|\lambda|$ percentiles (§D, Table D3).

1.3.18 C.18 Sectional sign-prior CCI

A sectional CCI basis with one knot per period (1982/1990/1993/2007, plus 2014/2017/2020/2021 extensions) placebo-tested on the back-extended sample: the sectional canonical (adj-R² 0.6805, $|\lambda|$ 0.2244) sits at the 37th adjusted-R² percentile and 60th $|\lambda|$ percentile — below the random median on fit, between the literal 4-knot and maximal-GETS constructions on mean reversion. On the modern 1988Q3+ sample the sectional reduction retains two survivors against the deployed protocol's four, fitting somewhat worse (adj-R² 0.726 vs 0.754; $\lambda = -0.203$ vs -0.246) with a 0.69 correlation between the two indices — a related but coarser credit signal (§D.4).

1.3.19 C.19 Two-equation SUR (consumption + house prices)

Joint SUR estimation of the consumption equation and a house-price ECM on the back-extended 1976Q3+ sample yields a residual correlation $\hat{\rho}(\varepsilon_C, \varepsilon_H) = -0.0109$ under equation-by-equation OLS and -0.0133 under SUR — negligible cross-equation linkage. The SUR coefficients move by under 5 per cent of themselves relative to OLS for every substantive term (10 per cent on the near-zero intercept); the consumption equation's λ shifts from -0.2101 to -0.2118 (+0.8 per cent) with SUR standard errors within 0.01 per cent of the OLS ones. Joint estimation gives no efficiency gain at the quarterly frequency; the case for the multi-equation framework rests on cross-equation parameter restrictions, not residual covariance (§C.2, §D.5).

1.3.20 C.20 Three-equation joint cross-equation CCI identification

See §D.5, Table D4, for the full joint-identification exercise (cross-equation sign-survival across the consumption, house-price, mortgage-stock and wealth equations).

1.4 D. Credit-conditions-index construction and placebo battery

1.4.1 D.1 The Muellbauer–Williams smoothed-step spline approach

Muellbauer and Williams (2012) construct CCI as a spline of SDMMMA smoothed-step dummies (a five-quarter moving average of a four-quarter moving average of a 0/1 step) at institutional turning points in the Australian financial-policy chronology, with each knot's coefficient constrained by a sign prior derived from institutional history (deregulation episodes positive, retrenchment episodes negative), enforced by Hendry–Krolzig (2005) drop-on-violation general-to-specific reduction. Williams' canonical paper uses four knots: 1979Q1 (Campbell Committee, removal of interest-rate ceilings), 1992Q1 (NBFI distress after the early-1990s recession), 1998Q1 (NBFI/securitisation expansion), and 2007Q1 (GFC retrenchment).

Table D1 — Direct replication of Williams' four canonical knots on the 1988Q3-onward sample.

Williams knot	Sign prior	Status on 1988+ sample
1979Q1	+	aliased (constant within window)
1992Q1	–	sign violator (data signal +ve)
1998Q1	+	sign violator (data signal –ve)
2007Q1	–	survives, coef ≈ -0.010

Only one of Williams' four canonical knots survives sign-prior reduction on this sample. The 1979 deregulation knot is mechanically uninformative because the smoothed step reaches unity by 1980Q2, three years before the estimation window opens; the 1992 and 1998 knots fail their institutional sign priors because the post-1988 sample observes the recovery from the early-1990s banking distress and the late-1990s NBFI period without the contrast against the prior tight regime that would identify the loosening direction. This is a direct consequence of the data window: ABS sectoral balance-sheet data begin only in 1988Q3, so the financial-liberalisation episode that most cleanly identifies the credit channels largely predates the sample on which the single-equation model is estimated. A direct 4-knot replication is therefore identifying one episode — the 2007 GFC tightening — plus a constant, not the four distinct episodes Williams' framework attributes to the spline.

1.4.2 D.2 The maximal-GETS Australian CCI (deployed construction)

Rather than impose Williams' published knot count, a maximal-GETS approach starts from a richer 15-knot candidate set covering the documented Australian financial-policy chronology (Campbell '79, housing-finance deregulation '86, state-bank distress '90, banking distress '92/'93, the Wallis report/APRA establishment '98, the GFC tightening '07, the deposit-guarantee scheme '08, the FHB Boost '09, the APRA macroprudential rounds '14/'17, the Hayne Royal Commission '19, the APRA cap removal/buffer reduction '19Q3, the COVID/JobKeeper episode '20, and the APRA buffer hike '21) and lets drop-on-violation reduction prune knots that are aliased or violate their institutional sign prior.

Table D2 — Surviving knots on the 1988Q3–2024Q4 sample under iterated drop-on-violation reduction (*cci_williams*).

Knot	Sign prior	Coef (OLS)	Reading
2007Q3	–	–0.0022	GFC tightening onset
2009Q1	+	+0.0123	First Home Buyer Boost (contestable, see below)
2019Q1	–	–0.0338	Hayne Royal Commission lending crackdown
2020Q2	+	+0.0071	COVID/JobKeeper income support

Nine candidate knots (1990Q3, 1992Q1, 1993Q1, 1998Q3, 2008Q4, 2014Q4, 2017Q1, 2019Q3, 2021Q4) violate their institutional sign priors and are dropped; 1979Q1 and 1986Q1 are aliased (constant within the window). The *cci_williams* series is built from these four surviving knots, peak-normalised to unity: identically zero from 1976 until 2007Q3, dips to ≈ -0.15 by late 2008, rises to its peak of 1 by 2010Q4, plateaus through 2018Q4, then falls to a trough of -2.12 in 2020Q4 and settles at ≈ -1.63 from 2022 onward (range -2.12 to $+1.00$).

Four implications follow. First, every credit channel in Specs 8 and 11 is identified off roughly 70 post-2007 quarters — before 2007Q4 each CCI interaction is exactly zero. Second, that all four surviving knots are post-2007 is itself part of the identification story (§D.5): the post-1988 sample carries usable sign-identifying variation only around the GFC, macroprudential and pandemic episodes. Third, the institutional reading of the surviving 2009Q1 "+" knot is contestable: the First Home Buyer Boost was a fiscal stimulus rather than a lending-standards easing, and the RBA's Financial Stability Reviews record lending standards tightening through 2009, so the "+" prior records a credit-demand event under a credit-supply label. Fourth, the candidate basis embeds a documented sign-prior conflict at 1993Q1: the maximal basis codes it -1 (the tail of the early-1990s bank retrenchment) while the sectional basis (§D.4) codes the same date $+1$ (mortgage-originator entry, Aussie Home Loans); both institutional readings are defensible for different events at the same date and the conflict is disclosed rather than silently resolved.

The reduction protocol requires precise statement: at each pass the consumption ECM is re-fitted with the currently surviving candidate set and all knots whose coefficient violates its sign prior are dropped simultaneously, iterating to a fixed point. This differs from Williams' one-at-a-time, strongest-violator-first reduction; the survivor set is protocol-dependent — a single-pass reduction over the same 15-knot basis retains a different set of five knots (1992Q1, 2007Q3, 2017Q1, 2019Q1, 2020Q2, with 2009Q1 aliased rather than surviving) and a different λ . The iterated protocol is the one deployed, and it is the one placebo-tested "as deployed" in §D.3.

The construction is two-step, with pre-test re-use of the dependent variable: the knots are first selected as plain additive long-run regressors inside a constant-MPC consumption equation estimated on the same $\Delta \ln c$ series; the surviving combination, peak-normalised, is then re-deployed multiplicatively (as *ha_x_cci*, *r_x_cci*, *yp_x_cci*, *hp_x_1_minus_cci* and the *cci_williams* level) in Specs 8 and 11. Spec 11's fit statistics are therefore conditional on a CCI that was pre-fitted, under sign priors, to the same dependent variable — a pre-test problem that the placebo battery (§D.3) quantifies but does not remove.

The maximal-GETS construction is defensible on two grounds: (i) the candidate set comes from documented Australian institutional history, not authorial choice of specific dates; and (ii) the surviving knots are those whose data signal aligns with their institutional sign prior, so the spline is empirically selected rather than imposed. It does not, however, deliver a structurally identified credit-conditions factor on its own — see §D.5 and the interaction-collinearity result in §C.4/§C.20.

1.4.3 D.3 The placebo battery

Whether the spline identifies genuine credit-conditions turning points — rather than acting as flexible detrending of the consumption- equation residual — is tested with random-knot placebos: 200 draws of knot dates, compared like-for-like with the institutional construction under the *same* protocol. For the literal-Williams comparison all four drawn knots are entered unconditionally (pure fit, no sign-prior reduction). For the maximal-GETS comparison each draw of 15 candidate knots passes through a single sign-prior reduction pass. The **deployed-protocol placebo** replaces the 15 institutional knot dates with random dates carrying the same sign-prior pattern and runs exactly the iterated drop-on-violation reduction of §D.2, so the deployed `cci_williams` construction is placebo-tested as deployed, pre-test step included.

Table D3 — Four placebo runs.

Construction	Protocol	Sample	adj R ² %ile	$ \lambda $ %ile	Verdict
Literal Williams 4-knot	unconditional 4-knot	1988Q3+ (n=146)	45th	56th	below R ² median — detrending critique vindicated
Literal Williams 4-knot	unconditional 4-knot	1976Q3+ (n=190)	36th	26th	below median on both — critique persists
Maximal-GETS canonical	single-pass reduce	1976Q3+ (n=190)	48th	70th	below R ² median — critique persists
Deployed <code>cci_williams</code>	iterated reduce	1988Q3+ (n=146)	84th	80th	moderate support

The literal Williams 4-knot construction sits at or below the placebo median on both samples (45th adjusted-R² percentile on the modern sample, 36th/26th on the extended sample); Williams' specific published-knot dates, entered as published, do not outperform random dates, and single-pass maximal-GETS reduction on the extended sample does no better (48th R² percentile, retaining fewer knots than the random median). The deployed construction fares better: under the iterated reduction actually used to build `cci_williams`, the institutional knot dates beat 84 per cent of random draws on adjusted R² and 80 per cent on $|\lambda|$, while retaining fewer knots (4) than the placebo median (5) — more fit with less flexibility than typical random constructions. This is moderate, not strong, support: one in six random draws still matches the deployed fit, the percentile is specific to the iterated protocol, and the construction re-uses the dependent variable. The standalone spline remains, at best, weakly distinguished

from flexible detrending; it is not on its own a structurally identified common factor. (For the deployed *Spec-11* multiplicative construction, carried through the full model rather than the additive selection stage alone, see the deployed-protocol Spec-11 placebo in Table F6.)

1.4.4 D.4 Sectional sign priors (period-based) tested

Williams' Australian-paper §5.1 specification imposes sign priors over periods rather than knot by knot:

Period	Sign prior	Rationale
1982–1990	non-negative	Financial deregulation
Early 1990s	non-positive	Banking sector distress
Mid-1990s–2006	non-negative	New entrants, securitisation
2007+	non-positive	GFC

A parallel CCI basis with one knot per period, extending Williams' four periods to cover post-2008 events (APRA 2014, APRA 2017, COVID 2020, APRA 2021), placebo-tested on the back-extended sample with 200 draws of eight random knots and eight random ± 1 priors, sits at the 37th adjusted- R^2 percentile and 60th $|\lambda|$ percentile ("detrending critique persists – sectional below random median"). Williams' specific period dating does not outperform random period placements on the post-deregulation-extended window. In the side-by-side coefficient comparison the sectional basis retains only two survivors against the maximal basis's four, with a weaker fit (adj- R^2 0.726 vs 0.754; $\lambda = -0.203$ vs -0.246) and a 0.69 correlation between the two indices – the period-prior construction delivers a related but coarser credit signal than the maximal-GETS reduction.

Overall placebo verdict. Neither the literal 4-knot construction, nor a single-pass maximal-GETS reduction, nor sectional sign priors delivers placebo support on the extended sample – the institutionally dated knots per se do not beat random dates. The one construction that does beat most random draws is the deployed iterated protocol on the modern sample, and even that support is moderate and conditional on the protocol. The CCI's standalone identification remains closer to a single-equation OLS using flexible smoothed-step dummies than to a structurally identified common factor.

1.4.5 D.5 Why the spline alone cannot identify the CCI as a common factor

Williams (2010) and Duca and Muellbauer (2013, ECB WP 1581) estimate the CCI inside a multi-equation system (consumption, house prices, mortgage stock, home-equity withdrawal – Williams; consumption plus a mortgage-refinancing rate via a Kalman-filter state-space model – Duca and Muellbauer) where the same latent credit variable enters all equations simultaneously, with a normalisation ($\zeta_h = 1$ in the house-price equation) identifying ζ_c , ζ_m , ζ_w as relative scalings. This cross-equation parameter restriction is what identifies the CCI as a common factor rather than as an equation-specific residual. In a single-equation OLS estimation the spline is fit only to the consumption-equation residual, with no constraint that the same knot loadings satisfy sign priors in the house-price, mortgage-stock or home-equity-withdrawal equations simultaneously.

Table D4 – Cross-equation joint sign-survival. The Williams 15-knot maximal candidate set is refit with the consumption equation, the house-price equation, the mortgage-stock

equation and a home-equity-withdrawal proxy equation simultaneously, requiring each knot to satisfy its institutional sign prior in every equation in which it is estimable.

Survival regime	Surviving knots	n
Consumption equation only (extended sample)	1979, 1986, 1992, 2007Q3, 2017Q1, 2020Q2	6
Joint 3-equation (C $\cap$ H $\cap$ M)	1986, 2017Q1	2
Joint 4-equation (C $\cap$ H $\cap$ M $\cap$ HEW)	1986	1

Of the six knots that pass their sign prior when fitted to consumption alone, only 1986 (financial deregulation) and 2017Q1 (APRA macroprudential round II) have signs consistent with their institutional priors across consumption, house prices and mortgage stock simultaneously, and only 1986 also survives the home-equity-withdrawal equation. The consumption-only identification is therefore overstated: four of the six consumption-passing knots are consumption-equation-specific and do not survive a cross-equation common-factor restriction.

Table D5 — House-price-equation CCI loading under joint identification.

HP equation, CCI loading	(a) cons-only CCI	(b) joint OLS	(c) joint SUR
Estimate	-0.016	+0.024	+0.024
Sign	× violator	✓	✓

Rebuilding `cci_williams_joint` from the two knots surviving the three-equation test flips the house-price equation's CCI loading from negative (consumption-only CCI) to positive (joint identification) — Williams' cross-equation sign restriction working as intended. The mortgage-stock equation's loading remains negative under this joint-sign-survival approximation (which weights surviving knots by consumption-equation coefficients); full FIML with parameter restrictions across all four equations would be needed to flip it — the exercise is illustrative of the mechanism, not a substitute for the FIML estimate. A complementary two-equation SUR (consumption + house prices) finds a negligible cross-equation residual correlation ($\hat{\rho} = -0.0109$ OLS, -0.0133 SUR — see §C.19): the case for joint estimation rests entirely on cross-equation parameter restrictions, not on residual-covariance efficiency gains.

Interaction collinearity. In the faithful LIVES form, CCI multiplies six channels jointly inside the long-run bracket, so the regressors that carry those channels are each approximately proportional to CCI over the sample: the five CCI-carrying regressors (`cci_williams`, `ha_x_cci`, `hp_x_1_minus_cci`, `r_x_cci`, `yp_x_cci`) have absolute pairwise correlations between 0.66 and 0.97, with the extremes $\rho(\text{cci_williams}, \text{yp_x_cci}) = -0.967$, $\rho(\text{cci_williams}, \text{ha_x_cci}) = +0.890$, and the weakest pair (`hp_x_1_minus_cci`, `r_x_cci`) still at 0.66. Five near-collinear regressors carrying distinct structural meanings cannot be separately free-estimated off a single equation: OLS allocates a near-singular design among them more or less arbitrarily, producing wrong-signed and insignificant individual loadings even when the joint contribution of the credit block is real — consistent with the sign failures of Spec 8 (§C.4), the identification-vs-detrending reallocation when the CCI is added to the baseline (permanent-income loading +51 per cent, speed of adjustment -92 per cent, net-liquid loading

+303 per cent, superannuation loading +258 per cent moving from Spec 6 to Spec 8), and the calibration collapse under Specs 10/12 (§B.5).

1.5 E. Permanent-income construction

Permanent income y_t^p is the forward-looking object in the ratio $\log(y_t^p/y_t)$, constructed as a discounted weighted average of *expected* future log income over a 40-quarter horizon ($k = 40$ quarters) at quarterly discount factor $\delta_q = 0.95^{1/4}$ (annual discount factor $\delta = 0.95$, annual discount rate $\eta = 0.05$):

$$\log(y_t^p/y_t) = E_t \left[\sum_{h=1}^{40} w_h \log(y_{t+h}) \right] - \log(y_t), \quad w_h = \frac{\delta_q^{h-1}}{\sum_{h=1}^{40} \delta_q^{h-1}}.$$

Two forecasters are implemented.

Italy direct forecast (headline measure). Following the Italian implementation's direct (single-regression) forecast of the discounted future-income aggregate (De Bonis, Liberati, Muellbauer and Rondinelli 2020, Appendix A.2), the discounted weighted average of future log income is regressed, in a single full-sample equation, on $\log(\text{lf_share})$ (the labour-force-participation term capturing slow-moving demographic effects on trend income), a trend, a post-2008 split trend, the four-quarter moving average of log income, the unemployment rate, and four-quarter-difference dynamics; the permanent-income series is the fitted value. Three properties are disclosed:

1. *Look-ahead and tail extrapolation.* The coefficients are estimated over the whole sample, so y_t^p embeds information dated after t and is non-causal — a two-sided measurement rather than a real-time forecast. The realised 40-quarter-ahead target is computable only up to 2014Q4, so the training sample ends there and the final forty quarters of y^p — about 27 per cent of the estimation sample, including the entire COVID period — are out-of-training extrapolations from 2014-vintage coefficients, driven mainly by the deterministic trend terms.
2. *GFC learning ogive.* The series is multiplied by an ogive that declines from 1 to 0.5 over 2008Q3–2012Q2, so the headline regressor is half the raw discounted-gap measure over the post-2012 two-thirds of the sample. A no-ogive re-estimate of the headline specification (§C.10) leaves the structural conclusions unchanged.
3. *Forecaster choice and the Australian permanent-income puzzle.* Under a rolling AR(8) forecaster the $\log(y^p/y)$ coefficient is negative (the "Australian permanent-income puzzle"); under the Italy-style direct forecaster it is positive, and the Italy forecaster fits better (AR $\text{adj-}R^2 = 0.696$, Italy $\text{adj-}R^2 = 0.731$, Spec 3 frame).

Rolling AR(8) (robustness). A rolling AR(8) regression of log income on eight own lags plus a linear trend, a post-2008Q3 step dummy and a trend-break interaction; forecasts are aggregated over the 40 horizons using the discount weights, with optional `unemp_rate`, `log_oil`, `log_reer`, `log_stocks` predictors and the same 2008Q3 ogive.

1.5.1 E.1 The real-time vs full-sample distinction

Table E1 — Permanent-income method comparison, two specification frames.

	PI measure	Frame	λ (ecm_lag)	t	$\log(\hat{y}^p/y)$	t	adj-R ²
AR	(expanding-win- dow)	Spec 3, n=146	-0.0479	-1.12	-0.0036	-3.96	0.696
	Italy LP (full-sample)	Spec 3, n=146	-0.1906	-2.90	+0.1963	+3.41	0.731
AR	(expanding-win- dow)	Spec 6, n=86	-0.0948	-1.68	-0.1583	-1.68	—
	Italy LP (full-sample)	Spec 6, n=86	-0.2386	-2.55	+0.3253	+1.50	—
	Italy LP (real-time)	Spec 6, n=86	-0.1594	-2.39	-0.1451	-2.23	—

The full-sample Italy LP measure is the headline measure (framed as a measurement, not a forecast); the AR forecaster delivers the negative "Australian permanent-income puzzle" coefficient in both frames (significant in the Spec 3 frame, $t = -3.96$). The real-time Italy LP variant is causal — re-fitting the projection at each t on data whose full k -quarter horizon is realised by t — and shows that roughly two-thirds of the full-sample Italy $|\lambda|$ is genuine (-0.159 against -0.239 , itself significant at $t = -2.39$; the look-ahead accounts for about a third), but that the **positive permanent-income sign is not**: it reverses to -0.145 ($t = -2.23$) under the causal measure. The full-sample measure is carried as the headline and its positive-PI sign and part of its λ magnitude are disclosed as full-sample, non-causal properties; the same look-ahead caveat attaches to the strong permanent-income coefficient in the faithful LIVES specification (Spec 11, $\ln_{yp_over_y} +0.459$, $t = 4.0$) — the real-time column is the operational robustness check, and any forecasting use of the equation (e.g. embedding it in a policy model) requires the real-time variant or the AR forecaster, not the full-sample measure.

Structural reasons the AR and Italy measures diverge: the rolling AR(8) forecaster lacks the labour-force-share predictor that captures Australia's slow-moving demographic effects on trend income, compounds short-run AR misspecification across 40 horizons, and over-estimates persistence after large income shocks, all of which the one-step direct projection avoids. The puzzle's reversal under the real-time Italy measure is read as a feature of full-sample permanent-income *measurement* rather than a clean real-time resolution.

1.5.2 E.2 Discount and horizon sensitivity

The permanent-income discount and horizon settings are not load-bearing for the speed of adjustment: across $\delta \in \{0.90, 0.95, 0.97\}$, $k \in \{20, 40, 60\}$ and the ogive on/off toggle, λ in the AR/Spec 2 frame moves only at the third decimal (full grid in §C.10). The forecaster *method* (Italy vs AR, full-sample vs real-time), not the discount calibration, is the material choice.

1.5.3 E.3 The structural gearing bound

In the faithful specification the permanent-income channel is strong and correctly signed: $\ln(\hat{y}^p/y)$ enters with OLS level coefficient $+0.459$ ($t = 4.04$) full sample and $+0.298$ ($t = 5.81$) pre-COVID. By the structural-recovery rule ($\gamma = \text{OLS}/|\lambda|$) the implied gearing is $\text{OLS}/|\lambda| \approx 1.02$ (full) to 1.13 (COVID-controlled) — **above the theoretical admissibility bound $\psi \leq 1 - \eta \approx 0.95$** implied by the discounting that defines y^p . The breach survives removing the ogive (structural ≈ 1.05 ; §C.10), and the delta-method interval $[0.86, 1.18]$ (Table F2) does not exclude 0.95 , so the violation is not itself statistically decisive. Candidate explanations: the unit-income restriction forcing $\log(\hat{y}^p/y)$ to absorb low-frequency consumption-to-income drift, and the

measure's non-causal, post-2014 tail-extrapolated construction (above) — under the causal real-time variant the coefficient reverses sign entirely (§E.1), so the strong positive gearing is a property of the full-sample *measurement*, not an operational forecasting relationship. Spec 11 does not separately estimate the ψ_0/ψ_1 split: the credit-gear component enters through a separate yp_x_cci interaction whose full-sample coefficient (-0.510 , $t = -1.51$) is insignificant and wrong-signed relative to Williams' calibrated $\psi_1 = +0.93$, though it turns right-signed ($+0.19$, $t = 1.80$) pre-COVID — the data identify the level of the permanent-income gearing but not its credit slope on a single equation (§B.5).

1.6 F. Structural-parameter inference

This section reproduces the original delta-method and residual- bootstrap confidence intervals on the implied structural coefficients (Tables F1–F2), and then reports four further exercises: a bounds/stationarity test of the long-run relationship (Tables F3–F4), a form-vs-sample ablation of the Spec 6 $\rightarrow$ Spec 11 gap (Table F5), a deployed-protocol placebo run on the full multiplicative Spec 11 construction (Table F6), and a nested bootstrap that re-estimates the CCI-knot selection and the permanent-income/ecm-lag construction inside each resample draw (Table F7). The nested-bootstrap intervals (Table F7) are the primary basis for structural-coefficient inference; Tables F1–F2 are retained for comparison only, since they hold the CCI and permanent-income constructions fixed and are shown in §F.4 to be anti-conservative by 5–190 $\times$ relative to the nested intervals.

1.6.1 F.1 Delta-method and bootstrap confidence intervals (comparison only)

The implied structural coefficients $\gamma_i = \beta_i/|\lambda|$ are ratios of two imprecisely estimated quantities. Delta-method standard errors are computed from the Newey–West covariance of (β_i, λ) — which carries the correlation between each numerator and the speed of adjustment — with a seeded moving-block residual bootstrap (block length 8, $B = 1000$) as a cross-check. Both hold the right-hand side fixed: they propagate sampling uncertainty in the ECM coefficients but not the first-stage uncertainty in the generated permanent-income and CCI regressors (both are held at their full-sample values in both methods), so the intervals below are a lower bound on the true uncertainty — the gap the nested bootstrap (§F.4) closes.

Table F1 — Spec 6 (conventional baseline, n=86).

Term	Implied γ	95% CI (delta)	Bootstrap median	Bootstrap 95% CI	Williams	Williams in CI?
Housing ha_y	0.009	[-0.056, 0.075]	0.008	[-0.049, 0.107]	0.0488	✓
Net liquid nla_y	0.035	[-0.262, 0.332]	0.027	[-0.183, 0.361]	0.159	✓
Equities eq_y	-0.065	[-0.468, 0.337]	-0.071	[-0.303, 0.208]	0.011	✓
Super super_y	0.025	[-0.040, 0.091]	0.027	[-0.064, 0.141]	0.011	✓
log(HP/y)	0.043	[-0.308, 0.393]	0.049	[-0.400, 0.325]	-0.130	✓
Permanent income	1.363	[0.285, 2.442]	1.377	[0.605, 2.192]	0.200	×
Wealth aggregate (Σ)	0.004	[-0.659, 0.668]	—	—	0.230	✓

Williams' Table 1 value lies inside the 95 per cent interval for every Spec 6 coefficient except the permanent-income gearing (whose interval excludes his calibrated 0.20 from above); the same intervals contain zero for every wealth channel and values far from Williams. On the conventional baseline the data cannot distinguish Williams' wealth profile from a broad range of alternatives, including no wealth effect at all — non-rejection driven by imprecision, not confirmation.

Table F2 — Spec 11 (faithful LIVES headline, n=146).

Term	Implied γ	95% CI (delta)	Bootstrap median	Bootstrap 95% CI	Williams	Williams in CI?
Housing $\times$ CCI ha_x_cci (γ_1)	0.0055	[-0.010, 0.021]	0.0052	[-0.0092, 0.0204]	0.0488	$\times$
Net liquid nla_y	0.060	[0.022, 0.098]	0.0609	[0.0129, 0.1107]	0.159	$\times$
Illiquid ilfa_y	0.035	[0.012, 0.057]	0.0347	[0.0047, 0.0647]	0.022	$\checkmark$
Affordabil- ity hp_x_1_minus_cci	0.062	[0.027, 0.098]	0.0622	[0.0328, 0.0973]	-0.130	$\times$
Permanent income (ψ_0)	1.024	[0.865, 1.184]	1.0221	[0.8988, 1.1615]	0.200	$\times$
PI $\times$ CCI yp_x_cci (ψ_1)	-1.138	[-2.099, -0.177]	-1.1374	[-1.5228, -0.7349]	0.930	$\times$
Wealth aggregate (Σ)	0.100	[0.036, 0.164]	—	—	0.230	$\times$

Under the fixed-regressor intervals, Williams' net-liquid m.p.c. (0.159), housing m.p.c. (0.0488), affordability sign, permanent-income gearing and credit slope would all lie outside the 95 per cent bands while his illiquid m.p.c. (0.022) would lie inside — but Table F7 shows none of these individual-channel comparisons survives once generated-regressor uncertainty is propagated, so no rejection of Williams' calibration is claimed on any channel. Agreement with Williams under this (fixed-regressor) inference is on form — the error-correction speed and the significance pattern of the wealth channels — not on the credit-channel calibrations. For use in a calibrated policy model, the aggregate long-run wealth coefficient is $\gamma_W = 0.100$ (delta CI [0.036, 0.164]): positive and bounded away from zero, but below both Williams' 0.230 and a calibrated net-wealth elasticity of ≈ 0.17 used in some Australian policy models (the latter sits just outside the upper bound). §F.4 shows that once the generated-regressor uncertainty is honestly carried, none of these individual-channel comparisons survives except the sign and non-zero magnitude of λ itself.

1.6.2 F.2 Bounds test and the imposed-ratio stationarity check

A Pesaran–Shin–Smith (PSS) bounds test on the unrestricted error-correction representation is used to establish whether a level (cointegrating) relationship exists, since the Engle–Granger residual test (Table B2/§B.2) fails to reject no-cointegration for every estimable specification.

Table F3 — PSS bounds test. Source: bounds-test output.

Spec	k	n	F-stat	I(0) bound 5%	I(1) bound 5%	t-stat	I(0) bound	I(1) bound	Verdict	
Spec 6 (con- ven- tional base- line)	6	9	86	5.591	2.14	3.30	-4.931	-2.86	-4.88	cointe- grated
Spec 11 (LIVES head- line)	11	9	145	9.572	2.14	3.30	-5.083	-2.86	-4.88	cointe- grated
Spec 12 (cali- brated)	12	—	—	—	—	—	—	—	—	infeasi- ble (see note)
Spec 12, free-re- gres- sors- only di- agnos- tic	12,	3	145	4.245	3.23	4.35	-4.002	-2.86	-3.78	incon- clusive

Both the F-statistic and the t -statistic clear their respective $I(1)$ upper bounds for Spec 6 ($F = 5.591 > 3.30$; $t = -4.931$ beyond -4.88) and Spec 11 ($F = 9.572 > 3.30$; $t = -5.083$ beyond -4.88): a level relationship is supported for both the conventional baseline and the headline specification. Spec 12's long-run vector is partly hard-calibrated ($\gamma_{\text{IFA}} = 0.022$, $\psi_0 = 0.20$, $\psi_1 = 0.93$ enter as a fixed, non-estimated offset inside an iteratively solved fixed point, not as freely estimated coefficients — the same reason the Engle–Granger screen skips Specs 10/12, §B.2), so the PSS unrestricted-ECM is not well-defined for Spec 12's actual estimating equation; the primary row is recorded infeasible. A diagnostic restricted to the two freely estimated long-run regressors only (ha_x_cci , nla_y , plus income) is inconclusive ($F = 4.245$, between the 3.23/4.35 bounds) and excludes the calibrated $\text{ilfa_y}/\text{ln_yp_over_y}/\text{yp_x_cci}$ offset, so it understates rather than tests Spec 12's true long run. Spec 12's long run must therefore be described as imposed/calibrated, not tested.

Table F4 — Stationarity of the imposed unit-income ratio $\text{ecm_lag} = \ln(\text{cons}_{\{t-1\}}) - \ln(\text{income}_t)$. Source: stationarity test output (ADF with drift; KPSS level test).

Window	n	ADF stat	ADF 5% CV	ADF verdict	KPSS stat	KPSS 5% CV	KPSS verdict
Full	193	-2.327	-2.88	nonstationary	0.997	0.463	nonstationary
Spec window 6	86	-2.428	-2.89	nonstationary	0.849	0.463	nonstationary
Spec window 11	146	-2.500	-2.88	nonstationary	0.783	0.463	nonstationary
Spec window 12	146	-2.500	-2.88	nonstationary	0.783	0.463	nonstationary

ADF and KPSS agree unanimously that the imposed unit-income consumption/income ratio is nonstationary in every window (ADF fails to reject the unit-root null in all four windows; KPSS rejects the stationarity null in all four). The level (cointegrating) relationship established by Table F3 is therefore carried by the **full** long-run regressor set — wealth, housing, credit conditions and permanent income acting jointly — and **not** by the imposed unit-income consumption/income ratio taken on its own.

1.6.3 F.3 Form vs sample — ablation of the Spec 6 → Spec 11 gap

The λ gap between the conventional baseline (Spec 6, $\lambda = -0.239$, $n = 86$) and the headline (Spec 11, $\lambda = -0.448$, $n = 146$) mixes two changes: functional form and sample/CCI-series extension. A 2×2 ablation swaps one factor at a time.

Table F5 — Form-vs-sample ablation of λ . Source: ablation output. Baseline reproductions match the Spec 6 and Spec 11 estimates of Tables B6 and B4 exactly.

Cell	Regress- sor form	Esti- mation sample	n	λ	λ t-stat	nla_y (t)	ilfa_y (t)	ilfyp_over_y (t)
Baseline	Spec 6 form sample (2002Q3– 2024Q4)	Spec 6	86	-0.2386	-2.552	0.0083 (0.24)	—	0.3253 (1.50)
Baseline	Spec 11 form sample (1988Q3– 2024Q4)	Spec 11	146	-0.4483	-3.567	0.0269 (3.75)	0.0155 (3.09)	0.4591 (4.04)
Cell A — form only	Spec 11 form sample (2002Q3– 2024Q4)	Spec 6	86	-0.5423	-3.807	0.0513 (1.45)	0.0206 (2.09)	0.4893 (3.74)
Cell B — sam- ple/ CCI- series only	Spec 6 form sample (1988Q3– 2024Q4)	Spec 11	146	-0.2617	-2.492	0.0192 (1.17)	—	0.2783 (2.05)

Decomposition of the total gap (Spec 6 baseline $\lambda = -0.239 \rightarrow$ Spec 11 baseline $\lambda = -0.448$; total gap -0.209): isolating functional form on Spec 6's own 86-quarter window (Cell A) moves λ to -0.542 — a larger movement than the entire gap to the headline estimate — whereas holding form fixed and only extending the sample and switching the credit-conditions series (Cell B) moves λ to just -0.262 , an order of magnitude smaller shift. The change in adjustment speed is driven by functional form, not by the larger sample.

Two qualifications are load-bearing. First, for the disaggregated wealth channels' *significance* — not their sign or rough magnitude — form and sample are complements, not substitutes: form alone (Cell A) gets the sign and rough magnitude of nla_y and $ilfa_y$ but does not reach conventional significance on $n = 86$ ($t = 1.45$ and 2.09) and only crosses significance once the sample is also extended (baseline Spec 11 $t = 3.75$ and 3.09); "form alone delivers significant disaggregated wealth channels" is not a supportable claim. Second, Cell B is form-*approximate*, not form-identical: $cci_williams$ spans negative values (a max-scaled spline index), so Spec 6's $\Delta^2 \log(CCI)$ short-run transform is undefined for it; a level second difference of $cci_williams$, lagged two quarters, is substituted as the closest available "acceleration in credit conditions" analogue, and the long-run block (nla_y , eq_y , $super_y$, ha_y , $ln_hp_over_y$, $real_rate$, $ln_yp_over_y$, $ln_yp_over_y_post2008$) is otherwise unchanged from Spec 6. Net permitted reading: **form is decisive for the adjustment speed; form alone does not activate the wealth channels' significance** — the two together do.

1.6.4 F.4 Deployed-protocol placebo on the full Spec-11 construction

The §D.3 placebo battery validates the additive knot-selection stage of the CCI (Table D3). A separate, deployed-protocol placebo carries each of 200 random-knot draws through the **full multiplicative Spec 11 construction** — the same object actually estimated, not just the additive selection stage — so that the placebo tests what is deployed, not what is merely selected.

Table F6 — Deployed-protocol Spec-11 placebo (198 finite draws of 200). Source: deployed-protocol placebo output.

Metric	Real (institutional) value	Placebo median	Placebo p10–p90	Real percentile	Higher is better?
adj-R ²	0.8244	0.7610	[0.7434, 0.8040]	93rd	yes
logLik	532.20	509.68	[504.49, 524.17]	93rd	yes
BIC	−954.75	−909.72	[−938.70, −899.34]	93rd	yes (lower BIC better)
$ \lambda $	0.4483	0.2402	[0.1974, 0.3794]	98th	yes
Joint Wald F, CCI block (5 terms)	6.88	2.32	[1.14, 4.97]	94th	yes

Meta: 200 knot draws targeted, 198 usable (finite) draws; the real institutional construction has 4 surviving knots from 15 candidates (additive stage), a joint-CCI-block Wald $p = 1.05 \times 10^{-5}$, and $\lambda = -0.4483$ on $n = 146$ (seed 20260611).

Carried through the full multiplicative Spec-11 construction, the institutionally timed credit-conditions index ranks at the 93rd percentile of the placebo distribution on model fit, at the 94th percentile on the joint significance of the credit-conditions block (Wald $F = 6.9$ vs a placebo median of 2.3), and at the 98th percentile on the estimated adjustment speed. The institutional knot placement is doing genuine identifying work at the deployed stage, not merely at the additive spline-selection stage of Table D3. This is **distributional (percentile) evidence that the CCI series as a whole is informative**; it is **not** evidence that any individual credit-interaction channel is separately identified — those are governed by Table F7. The 94th-percentile ranking of the CCI block does not license a claim that any single channel within it (e.g. housing $\times$ CCI, rate $\times$ CCI) is significant.

1.6.5 F.5 The nested bootstrap — headline structural-coefficient inference

Tables F1–F2 hold the CCI-knot selection and the permanent- income/ecm_lag construction fixed at their full-sample point estimates, so they understate the true sampling uncertainty in the structural coefficients (§F.1). A nested bootstrap ($B = 199$ usable draws) instead re-selects the CCI knot set (the additive drop-on- violation reduction of §D.2) **and** the ecm_lag/ permanent-income construction **inside** each resample draw, so both stages of generated-regressor uncertainty are carried through to the final coefficient interval.

Table F7 — Nested 95% confidence intervals, Spec 11 (headline). Source: nested-bootstrap output, $B = 199$.

Coefficient	Point	Conventional 95% CI	Nested median	Nested 95% CI	Nested excludes 0?	Williams	Williams in nested CI?
λ (adjustment speed)	-0.4483	[-0.6946, -0.2019]	-0.1686	[-0.2728, -0.0774]	YES	-0.286	—
nla_y (γ)	0.0600	[0.0224, 0.0975]	0.0381	[-0.1739, 0.3168]	no	0.159	TRUE
ilfa_y (γ)	0.0345	[0.0120, 0.0570]	0.0122	[-0.0857, 0.1643]	no	0.022	TRUE
ln_yp_over_y (γ)	1.0242	[0.8648, 1.1837]	0.6960	[-0.1238, 1.6014]	no	0.200	TRUE
yp_x_cci (γ)	-1.1379	[-2.0987, -0.1771]	-25.597	[-103.31, 24.46]	no	0.930	TRUE
ha_x_cci (γ)	0.0055	[-0.0104, 0.0214]	-0.2847	[-3.662, 2.357]	no	0.0488	TRUE
hp_x_1_minus_cci (γ)	0.0623	[0.0269, 0.0976]	0.0728	[-0.0737, 0.3755]	no	-0.130	FALSE

Width-inflation of the nested interval relative to the conventional (delta-method/NW) interval: $\lambda \times 0.40$ (nested is *narrower*, since the nested statistic is the raw coefficient rather than a ratio); the structural γ ratios inflate by $\times 5.4$ – $\times 6.5$ for nla_y , $ilfa_y$ and $ln_yp_over_y$, $\times 6.4$ for $hp_x_1_minus_cci$, and $\times 66$ – $\times 189$ for the two ratio-heavy CCI-interaction γ (yp_x_cci , ha_x_cci) – a mechanical consequence of dividing by a bootstrap-resampled λ whose distribution includes draws close to zero.

Binding reading. Only λ survives: its nested interval $[-0.273, -0.077]$ excludes zero, so the ECM adjustment speed's sign and non-zero magnitude are identified once knot-selection and ecm_lag /permanent-income construction uncertainty are carried honestly. Note the nested interval is centred well inside the conventional interval $([-0.695, -0.202])$ and the nested median $|\lambda| \approx 0.17$ is attenuated relative to the 0.45 point estimate – a mechanical consequence of letting ecm_lag become a bootstrap-generated regressor; λ 's sign and significance are the robust finding, with the level attenuation flagged. All six structural γ channels are statistically indistinguishable from zero once knot-selection and PI/ecm_lag construction uncertainty are honestly propagated – each is reported as sign-only, with magnitude unidentified. Williams' calibrated value lies inside the nested CI for nla_y , $ilfa_y$, $ln_yp_over_y$, yp_x_cci , and ha_x_cci ; for $hp_x_1_minus_cci$ Williams' -0.13 sits just outside the nested CI, but that channel is itself indistinguishable from zero, so no individual-channel "reject Williams" claim is available on any of the six structural elasticities. The delta-method intervals of Tables F1–F2 are therefore anti-conservative for the structural γ and are retained here as an appendix comparison only, not as headline inference.

1.6.6 Table and note index

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References. Citations in this appendix are as in the main-text reference list, plus: Foster, R. A. (1996). *Australian Economic Statistics 1949–50 to 1996–97*. Reserve Bank of Australia Occasional Paper No. 8. MacKinnon, J. G. (1991). Critical values for cointegration tests. In R. F. Engle and C. W. J. Granger (eds.), *Long-Run Economic Relationships: Readings in Cointegration*, 267–276. Oxford: Oxford University Press. MacKinnon, J. G. (2010). Critical values for cointegration tests. Queen's Economics Department Working Paper No. 1227.